ISO 50001 EnMS
Internal Audit Report No.1

EnMS System	ISO 50001:2018
Revision Number	1.0
Issue Date	22 Nov 2019

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1 Audit checklist

ISO clause no.	Description	Covered in this audit
4. Context of the organization		
4.1	Understanding the organization and its context	✓
4.2	Understanding the needs and expectations of interested parties	✓
4.3	Determining the scope of the energy management system	✓
4.4	Energy management system	✓
5. Leadership		
5.1	Leadership and commitment	✓
5.2	Energy policy	✓
5.3	Organizational roles, responsibilities and authorities	✓
	6. Planning	
6.1	Actions to address risks and opportunities	✓
6.2	Objectives, energy targets and planning to achieve them	✓
6.3	Energy review	✓
6.4	Energy performance indicator (EnPI)	✓
6.5	Energy baseline	✓
6.6	Planning for collection of energy data	
7. Support		
7.1	Resources	✓
7.2	Competence	✓
7.3	Awareness	✓
7.4	Communication	✓
7.5	Documented information	✓
8. Operation		
8.1	Operation planning and control	✓
8.2	Design	✓
8.3	Procurement	✓
9. Performance evaluation		
9.1	Monitoring, measurement, analysis and evaluation of energy performance	✓
9.2	Internal audit	✓
9.3	Management review	✓
10. Improvement		
10.1	Non-conformity, corrective action	✓
10.2	Continual improvement	✓

2 Audit finding

RECORD OF INTERNAL AUDIT: NONCONFORMITY / AREAS FOR IMPROVEMENT / POSITIVE FINDING

Organization: MGM COTAI	Auditor: Arlene Guo	Standard: 6.1.1, 6.1.2,6.2.2, 6.3
Date: 12 Nov 2019	Location of Finding: Energy Management System (EnMS) Manual	

Type of Finding (check one): ☐ Nonconformity ☒ Areas for Improvement ☐ Positive Finding

Facts Concerning the Finding*

Statement of Criteria (Requirement):

6.1 Actions to address risks and opportunities
6.2 Objectives, energy targets and planning to achieve them
6.3 Energy review

Objective Evidence (Statement of Finding):

The risks and opportunities on the EnMS Manual were yet to be approved by the top management / the management representative.

** Statement of Criteria (Requirement) is not required for positive findings, but should be provided if appropriate.*

Documentation Reference:

Energy Management System (EnMS) Manual

ISO Requirement Reference:

As per clause 6.1.1, 6.1.2,6.2.2, 6.3

Auditee Signature:	Auditor Signature

Auditee signature indicates that facts concerning the nonconformity are correct.

Response and follow up:
Update as policy and store in MGM internal web “DPO_FM_GNC_Pol_Cotai Energy Management System.docx”

Response person: MGM Cotai FM Kevin Pong	Date : 29/12/2019

RECORD OF INTERNAL AUDIT: NONCONFORMITY / AREAS FOR IMPROVEMENT / POSITIVE FINDING

Organization: MGM COTAI	Auditor: Arlene Guo	Standard: 6.2.2, 6.3
Date: 12 Nov 2019	Location of Finding: MGM Cotai Resource Efficiency Management Report	

Type of Finding (check one): ☐ Nonconformity ☒ Areas for Improvement ☐ Positive Finding

Facts Concerning the Finding*

Statement of Criteria (Requirement):

6.2 Objectives, energy targets and planning to achieve them
6.3 Energy review

Objective Evidence (Statement of Finding):

The Significant Energy Uses (SEUs) were not reported in monthly Resource Efficiency Management review.

It was recommended to consolidate the electricity & natural gas breakdown as an overall energy consumption breakdown as supporting of SEUs identification

** Statement of Criteria (Requirement) is not required for positive findings, but should be provided if appropriate.*

Documentation Reference:

MGM Cotai Resource Efficiency Management Report

ISO Requirement Reference:

As per clause 6.2.2, 6.3

Auditee Signature:	Auditor Signature

Auditee signature indicates that facts concerning the nonconformity are correct.

Response and follow up:
Energy Consultant Energenz updated the monthly energy report from December 2019 per finding recommendation.

Response person: Energenz	Date : 23/12/2019

RECORD OF INTERNAL AUDIT: NONCONFORMITY / AREAS FOR IMPROVEMENT / POSITIVE FINDING

Organization: MGM COTAI	Auditor: Arlene Guo	Standard: 4.1, 4.2, 9.3.1
Date: 12 Nov 2019	Location of Finding: Management Review	

Type of Finding (check one): ☐ Nonconformity ☒ Areas for Improvement ☐ Positive Finding

Facts Concerning the Finding*

Statement of Criteria (Requirement):

4.1 Understanding the organization and its context
4.2 Understanding the needs and expectations of interested parties
9.3 Management Review

Objective Evidence (Statement of Finding):

Management review was in the plan but the meeting and records were pending.

** Statement of Criteria (Requirement) is not required for positive findings, but should be provided if appropriate.*

Documentation Reference:

Management Review

ISO Requirement Reference:

As per 4.1, 4.2 , 9.3.1

Auditee Signature:	Auditor Signature

Auditee signature indicates that facts concerning the nonconformity are correct.

Response and follow up:
Management review conducted on 28/11/2019.

Response person: MGM Cotai FM / Energenz	Date : 28/11/2019

RECORD OF INTERNAL AUDIT: NONCONFORMITY / AREAS FOR IMPROVEMENT / POSITIVE FINDING

Organization: MGM COTAI	Auditor: Arlene Guo	Standard: 8.1, 8.3
Date: 12 Nov 2019	Location of Finding: Chiller Plant (Dafoo)	

Type of Finding (check one): ☐ Nonconformity ☒ Areas for Improvement ☐ Positive Finding

Facts Concerning the Finding*

Statement of Criteria (Requirement):

8.1 Operational planning and control
8.3 Procurement

Objective Evidence (Statement of Finding):

Non-Conformance Notice record was not provided regarding the abnormal findings indicated in Chiller Plant monthly report.

Evaluation of Completed Job was not indicated in Chiller Plant monthly report.

** Statement of Criteria (Requirement) is not required for positive findings, but should be provided if appropriate.*

Documentation Reference:

MGM_CCP Monthly report (Chiller Plant maintenance record)
Sustainable Procurement (Policy)

ISO Requirement Reference:

As per 8.1, 8.3

Auditee Signature:	Auditor Signature

Auditee signature indicates that facts concerning the nonconformity are correct.

Response and follow up:
MGM Cotai Plant team co-ordinated with vendor Dafoo to update the monthly service report from December 2019.

Response person: MGM Cotai FM Plant team	Date : 6/1/2020

RECORD OF INTERNAL AUDIT: NONCONFORMITY / AREAS FOR IMPROVEMENT / POSITIVE FINDING

Organization: MGM COTAI	Auditor: Arlene Guo	Standard: 9.1, 10.2
Date: 13 Nov 2019	Location of Finding: Façade Lighting Performance Control Panel	

Type of Finding (check one): ☐ Nonconformity ☐ Areas for Improvement ☒ Positive Finding

Facts Concerning the Finding*

Statement of Criteria (Requirement):

9.1 Monitoring, measurement, analysis and evaluation of energy performance and the EnMS
10.2 Continual Improvement

Objective Evidence (Statement of Finding):

Façade lighting was tuned down from the original level of 5000 lux to 2000 lux to reduce the building energy consumptions.

** Statement of Criteria (Requirement) is not required for positive findings, but should be provided if appropriate.*

Documentation Reference:

Façade Lighting daily checking log
Façade Lighting control panel training material
Façade Lighting luminosity standard
Procurement of façade lighting / LED display record

ISO Requirement Reference:

As per 9.1, 10.2

Auditee Signature:	Auditor Signature

Auditee signature indicates that facts concerning the nonconformity are correct.

Response and follow up:
MGM IT and MGM Cotai FM Tower Maintenance team provided those records.

Response person: MGM Cotai FM	Date : 28/12/2019.

RECORD OF INTERNAL AUDIT: NONCONFORMITY / AREAS FOR IMPROVEMENT / POSITIVE FINDING

Organization: MGM COTAI	Auditor: Arlene Guo	Standard: 9.1, 10.2
Date: 13 Nov 2019	Location of Finding: Theatre Back-Of-House area	

Type of Finding (check one): ☐ Nonconformity ☐ Areas for Improvement ☒ Positive Finding

Facts Concerning the Finding*

Statement of Criteria (Requirement):

9.1 Monitoring, measurement, analysis and evaluation of energy performance and the EnMS

10.2 Continual Improvement

Objective Evidence (Statement of Finding):

Energy saving mode is adopted in Theatre HVAC system.

Non-essential (safety and maintenance) theatre luminaires are dimmed by 70% - 90% to achieve energy savings.

** Statement of Criteria (Requirement) is not required for positive findings, but should be provided if appropriate.*

Documentation Reference:

Theatre HVAC, lighting and Gala Lift checking log
Theatre HVAC, lighting and Gala Lift maintenance record

ISO Requirement Reference:

As per 9.1, 10.2

Auditee Signature:	Auditor Signature

Auditee signature indicates that facts concerning the nonconformity are correct.

Response and follow up:
MGM Cotai FM Theatre team provided those records.

Response person: MGM Cotai FM Theatre Team	Date : 21/11/2019

3 Appendix - Internal Audit Summary

Date: 12-Nov-2019 and 13-Nov-2019
Time: 10:00-13:00, 14:00 to 16:30
Location: MGM Cotai Facilities Management Office
Auditor: MGM Cotai Sustainability (Bernard Cheng, Leona Cheang, Arlene Guo)
Auditee: MGM Cotai Facilities Management (Kevin Pong, Ann Leong)

I. Introduction

2 days meeting is called for 2019 “ISO50001:2018 Internal Audit” for MGM Cotai in compose of opening, document review, site investigation, reporting discussion and closing.

- Opening meeting: introduction of 2 days internal audit program

- Document reviewed (Clause 7.5):
  (1) Energy Management System (EnMS) Manual - (Clause 4.1 / 4.4 / 5.3)
  (2) Energy Management System (EnMS) Procedural Manual - (Clause 4.3/ 5.1 /5.2/ 6.3 /6.4 /6.6/ 7.4)
  (3) Risk and opportunities to EnMS - (Clause 4.2 / 6.1/ 6.2)
  (4) MGM Cotai Energy management opportunities (EMO) register - (Clause 6.2 / 6.5)
  (5) MGM Cotai Resource Efficiency Management Report - (Clause 8.1/9.1)
  (6) Power Management System Operation and Maintenance attendance (Training) - (Clause 7.2)
  (7) HV System Operation Training attendance (Training) - (Clause 7.2)
  (8) China Green Building (Design) application Self Assessment Report-MGMC (vBig5) - (Clause 8.2)
  (9) MGM Cotai – Lighting Load recommendation
  (10) Cotai Chiller Manual Staging (Policy)
  (11) MGM_CCP Monthly report (Chiller Plant maintenance record) - (Clause 8.1)
  (12) Sustainable Procurement (Policy) - (Clause 8.3)
  (13) MGM Cotai Monthly REM Workshop (Report) - (Clause 7.1 / 7.3)

- Site Inspection:
  Day One
  (1) 1F Chiller Plant
  (2) GF Platinum Kitchen

  Day Two
  (1) B1 Digital and Technology Office
  (2) B1 E-bus parking arear
  (3) Theatre BOH area

II. Interviewed Issues (Clause 9.2)

Day One

- Guest room (VDA system) and carpark lighting & ventilation are excluded in this internal audit, which may need to be scheduled for this can be potential area in external audit.
- Chiller Plant set points are checked in site inspection that are complied with the Cotai Chiller Manual Staging (Policy)

Day Two

- Façade Lighting Performance Control Panel
- Façade lighting has key down from the original design of 5000 lux to 2000 lux based on the concept of energy saving and environmental friendly
- Energy saving mode are adopted in Theatre HVAC and lighting system to save energy, lighting (except maintenance and safety needs) are dimmed to 10% - 30%